Industrial Internship Report on

”Portfolio”

Prepared by

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Executive Summary
This report presents the details of my industrial internship completed under Upskill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT). The internship duration was 4 weeks, during which I worked on developing a full-stack web application using React. The main objective of this project was to design and develop a responsive and interactive web application that can efficiently handle user interactions and display dynamic content. The project involved frontend development using React and integration of modern web technologies. During the internship, I gained practical knowledge of web development, component-based architecture, and real-world project implementation. This experience helped me enhance my technical skills and problem-solving abilities. Overall, the internship provided valuable industry exposure and hands-on learning experience.

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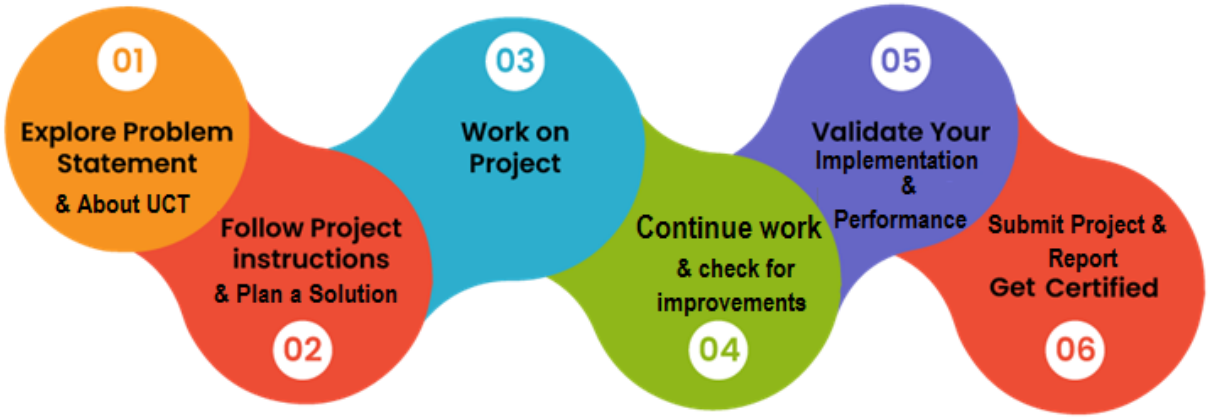
1 Preface

Internships play a crucial role in bridging the gap between academic learning and industry requirements. This internship provided me an opportunity to work on real-world projects and understand practical implementation of concepts.

During the internship, I worked on a full-stack React-based web application. The project helped me understand the workflow of web development, including designing user interfaces and managing data effectively.

I would like to express my sincere gratitude to Upskill Campus, UniConverge Technologies Pvt Ltd, and all mentors who guided me throughout the internship. Their support helped me complete the project successfully.

This internship has improved my technical skills, confidence, and ability to solve real-world problems. I hope this experience will help me in my future career growth.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

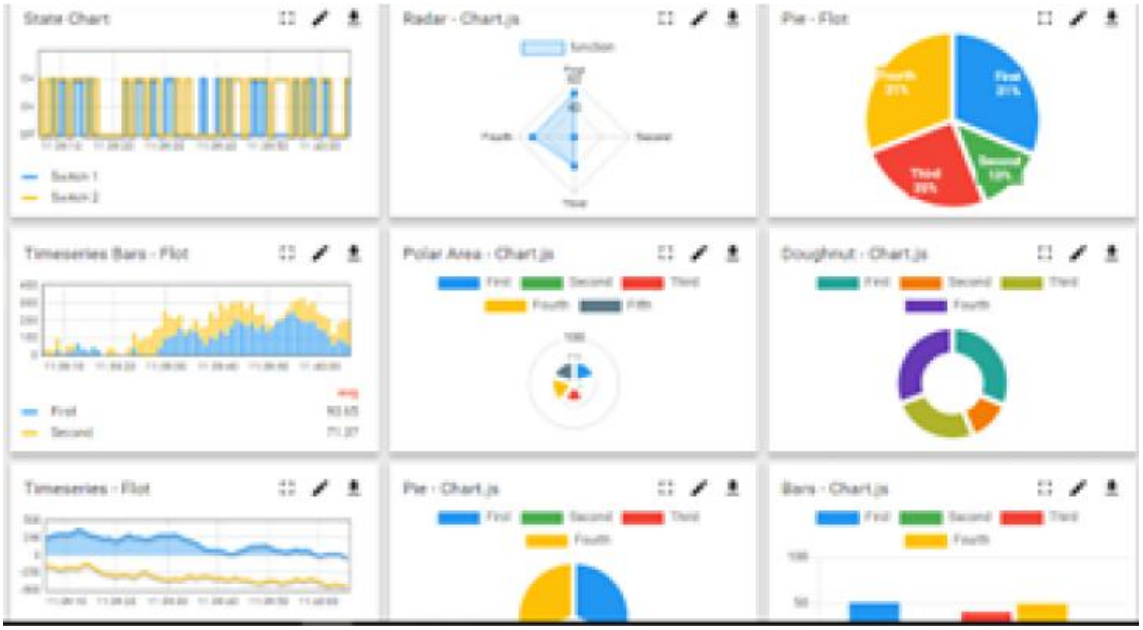
i. UCT IoT Platform (uct Insight)

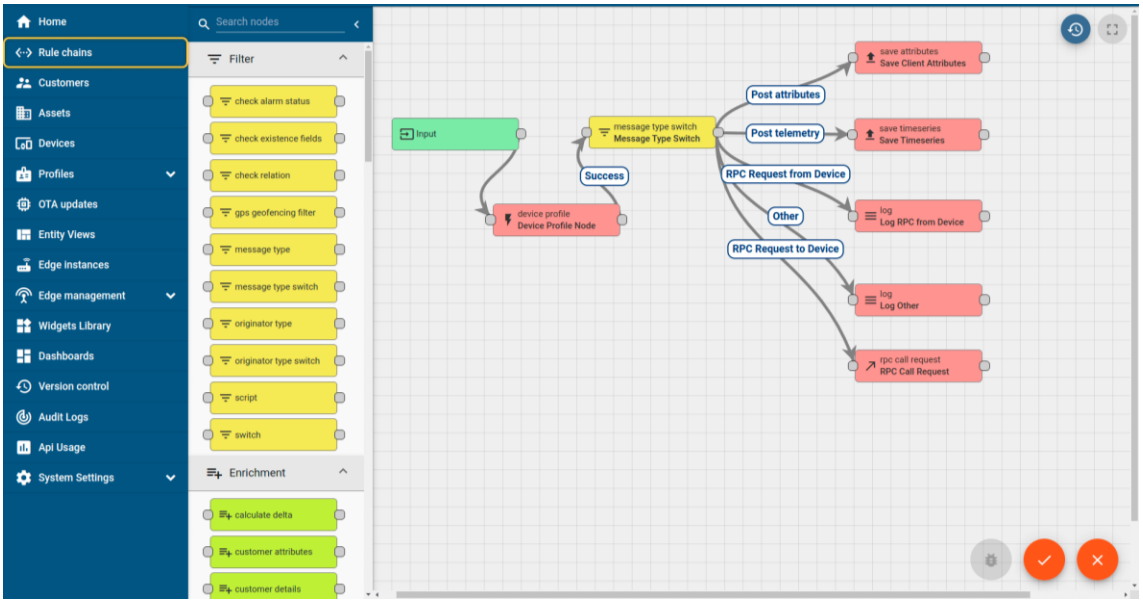
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

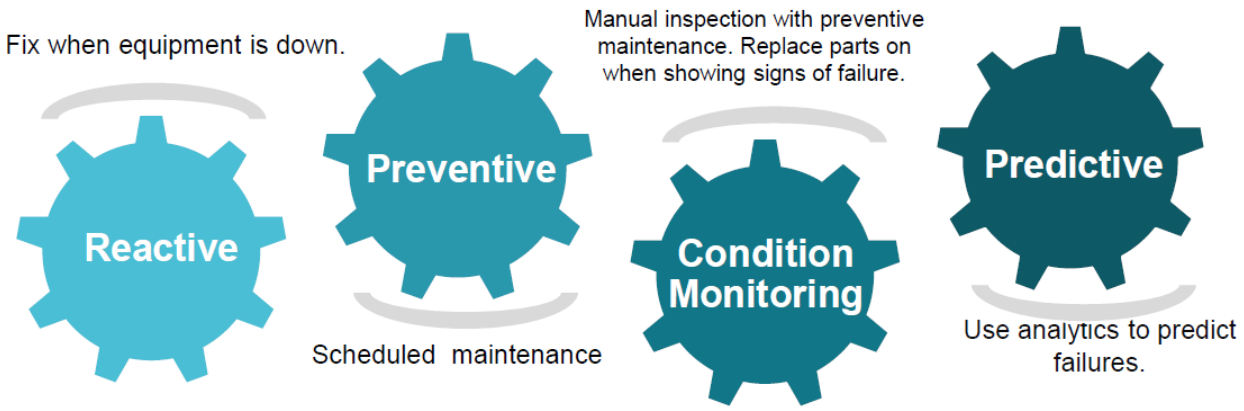
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

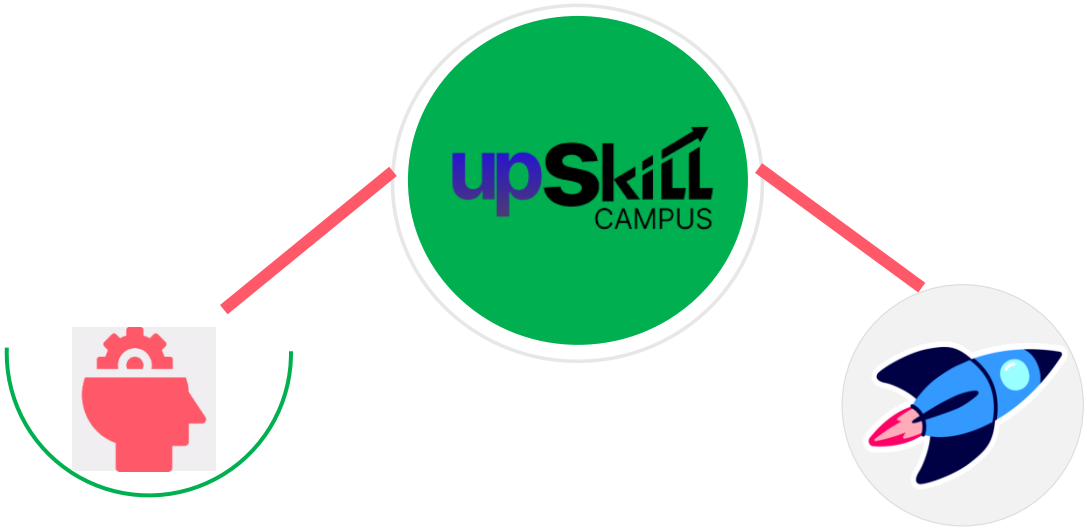
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

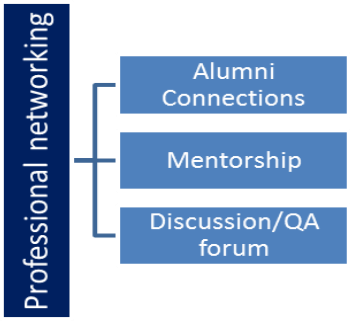
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

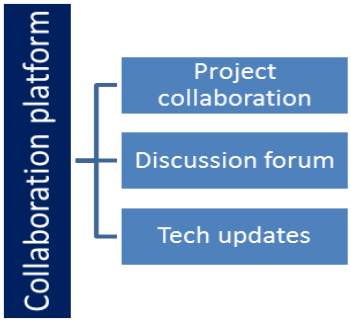


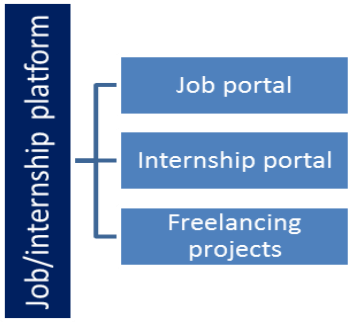
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

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2.6 Glossary

Terms	Acronym

3 Problem Statement

In the assigned problem statement

In today's digital world, having an interactive and responsive web application is essential for effective user engagement. Traditional static websites fail to provide dynamic content and real-time interaction.

The problem is to develop a modern web application that can efficiently handle user input, display dynamic content, and provide a seamless user experience. The application should be responsive, scalable, and easy to use.

The goal of this project is to design and implement a full-stack web application using React that solves these limitations and provides an improved user experience.

4 Existing and Proposed solution

Existing Solution:

Traditional websites are mostly static and lack interactivity. They do not provide dynamic content updates and often result in poor user experience. Managing such systems is also difficult and time-consuming.

Proposed Solution:

The proposed solution is a full-stack web application developed using React. It provides a dynamic and responsive user interface with better performance and usability. The application uses component-based architecture, making it scalable and easy to maintain.

Value Addition:

- Improved user experience

- Faster performance

- Responsive design

- Easy scalability

4.1 Code submission (Github link) : https://github.com/Tharun-575/upskillcampus

4.2 Report submission (Github link) : https://github.com/Tharun-575/upskillcampus/blob/main/Tharun_USC_UCT.pdf

5 Proposed Design/ Model

The system is designed using a component-based architecture where the application is divided into reusable components. Each component handles a specific functionality, making the system modular and efficient.

Workflow:

1. User interacts with the interface

2. React components process the input

3. Data is managed and updated dynamically

4. Output is displayed to the user

The design ensures smooth navigation, efficient data handling, and responsive layout across devices.

6 Performance Test

The application was tested for performance, responsiveness, and usability.

Test Cases:

- Loading of web pages

- User interaction with components

- Responsiveness on different screen sizes

Test Procedure:

The application was tested by running it in a browser and checking its behavior under different conditions.

Performance Outcome:

- Fast loading time

- Smooth user interaction

- Responsive design across devices

Limitations:

- Requires internet connection

- Can be improved with backend optimization

6.1 Test Plan/ Test Cases

The test plan is designed to evaluate the functionality, performance, and usability of the developed React web application. The testing ensures that all components work correctly and provide a smooth user experience.

Test Cases:

1. Application Loading Test

Input: Open the application in a web browser

Expected Output: Application should load without errors and display the homepage

2. Navigation Test

Input: Click on different sections (Home, About, Projects, Contact)

Expected Output: Smooth navigation between pages without delay

3. Responsive Design Test

Input: Open the application on different screen sizes (mobile, tablet, desktop)

Expected Output: Layout should adjust properly for all devices

4. User Interaction Test

Input: Click buttons, fill forms, and interact with UI elements

Expected Output: Proper response to user actions without errors

5. Error Handling Test

Input: Enter invalid data in input fields

Expected Output: Application should handle errors gracefully

6. Performance Test

Input: Load the application multiple times

Expected Output: Fast loading and consistent performance

The above test cases ensure that the application is reliable, user-friendly, and performs efficiently under different conditions.

6.2 Test Procedure

The testing of the application was carried out systematically to ensure that all functionalities work as expected. The following procedure was followed during testing:

1. Environment Setup

The application was executed in a web browser using a local development server. Required dependencies were installed using npm.

2. Application Execution

The project was run using the command "npm start" or "npm run dev", and the application was accessed through localhost.

3. Functional Testing

Each component of the application was tested individually to verify that it performs the intended function. Navigation, buttons, and user inputs were checked.

4. UI/UX Testing

The user interface was tested for responsiveness, layout consistency, and ease of use across different screen sizes and devices.

5. Error Testing

Invalid inputs and edge cases were tested to ensure the application handles errors properly without crashing.

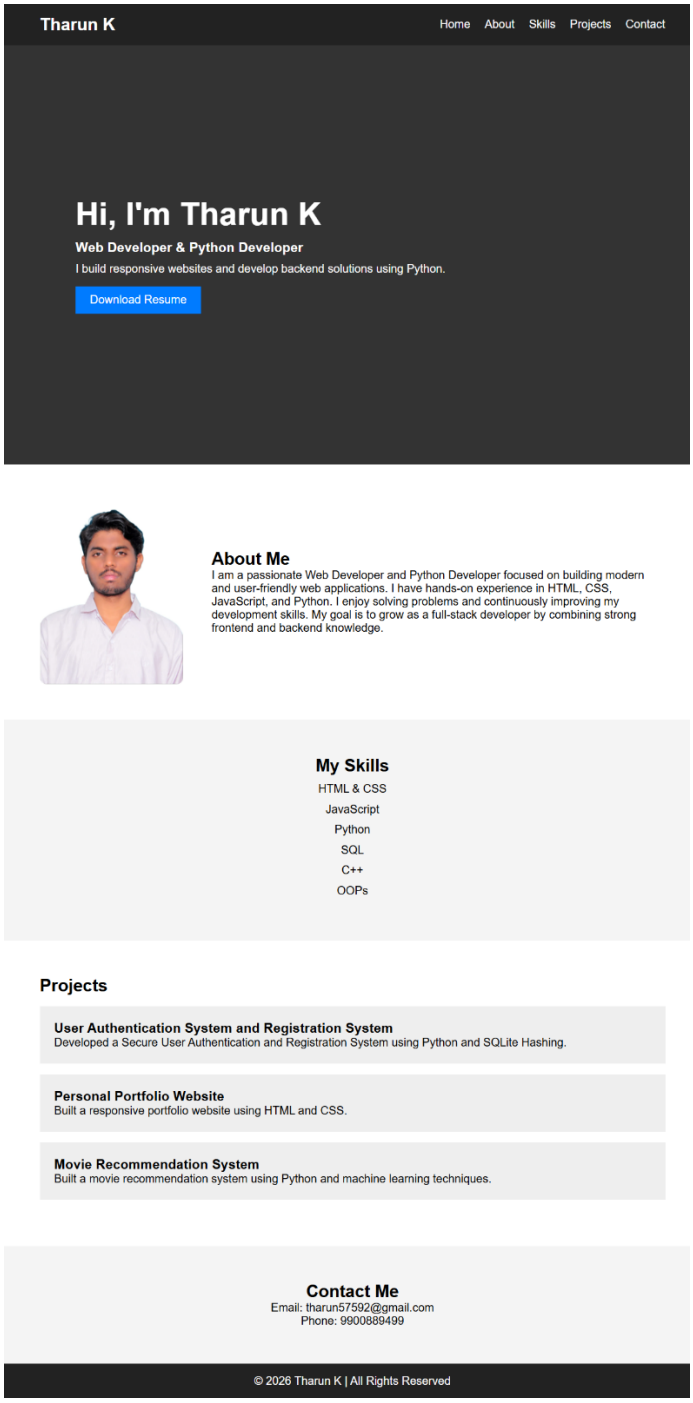
6. Performance Testing

The application was tested for loading speed and smooth interaction under normal usage conditions.

7. Final Validation

After testing all features, the application was reviewed to ensure that it meets the project requirements and works efficiently.

6.3 Performance Outcome



The performance of the developed React web application was evaluated based on speed, responsiveness, and user interaction.

The application demonstrated fast loading times when accessed through the browser. Navigation between different sections was smooth and did not cause any delays. The user interface responded quickly to user inputs such as button clicks and form submissions.

The application was also tested on different screen sizes, and it maintained a consistent and responsive layout across devices including mobile, tablet, and desktop.

During testing, the application handled most user interactions efficiently without any major errors. Minor limitations such as dependency on internet connection and scope for backend optimization were identified.

Overall, the performance of the application is satisfactory and meets the expected requirements for a modern web application.

7 My learnings

During this internship, I learned many technical and professional skills. I gained hands-on experience in React and understood how to build real-world web applications.

I learned about component-based architecture, state management, and handling user interactions. I also improved my knowledge of HTML, CSS, and JavaScript.

This internship enhanced my problem-solving skills and helped me understand how to debug errors and improve performance. I also learned how to manage projects and work independently.

In addition, I developed communication skills and gained confidence in presenting my work. This experience will help me in my future career as a software developer.

8 Future work scope

The project can be further improved by adding backend integration for storing data permanently. Additional features like authentication, database connectivity, and advanced UI enhancements can be implemented.

The application can also be converted into a mobile-friendly or cross-platform application. Future improvements can make the system more secure, scalable, and efficient..